

Authorship, Not Delay: What Frontier Distillation Buys an On-Device Small-Language-Model Traffic Controller

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Abstract—Large-language-model traffic signal controllers are evaluated in the cloud, but junctions run at the edge, and after an accident someone must prove which controller did what. We present, to our knowledge, the first traffic signal controller fine-tuned and served entirely on-device: a QLoRA-distilled small language model (SLM) compiled to int4 ONNX and served by Microsoft Foundry Local on a 6GB consumer GPU, controlling measured London topologies in SUMO behind a MaxPressure safety shield with a tamper-evident, Ed25519 hash-chained decision audit. Distilling a frontier teacher lifts held-out decision accuracy from 55–69% to 97–99% and saturates by 0.6B parameters: a 3.8B student is statistically indistinguishable from a 0.6B one, within $\pm 0.38\%$ of closed-loop delay. Whether accuracy becomes delay depends on whether the test network can express it: on saturated maps, over 30 training-disjoint seeds, no arm significantly beats a fixed-time floor. But on the same grid demand-calibrated so that traffic actually flows, the distilled controller beats that floor by 8.8% while the stock one loses to it by 11.9%, an 18.3% head-to-head improvement on 30 of 30 seeds ($p=2.0\times 10^{-21}$) that gridlock had been masking. The same calibration does not reproduce that effect on a second network, and we report that failed replication. Everywhere, fine-tuning raises the SLM’s share of logged decision ticks from 0.39–0.62 to 0.56–0.77, because the distilled policy trips the anti-starvation floor far less often, while its divergence from MaxPressure falls from 0.46–0.60 to 0.04–0.17. We argue authorship is the property an audit needs: an audit of the stock deployment could attribute two decisions in five to the learned component, against four in seven after distillation, a deterministic timer authoring the rest. We state the deflationary reading, that distillation partly taught imitation of the classical reference, and bound it with the evidence.

I. INTRODUCTION

LLM-based traffic signal control has moved quickly from prompting to fine-tuning [1]–[3], but the deployment story has not moved with it: every published controller is evaluated against a cloud endpoint, while the signal cabinet at a junction is an edge device with tight memory, no service-level guarantee to the cloud, and a statutory duty, unique among LLM application domains, to explain itself after a collision. A junction controller that cannot be audited is not deployable regardless of its delay numbers.

This paper asks what fine-tuning actually contributes to an SLM traffic controller that runs *entirely* on-device under a classical safety shield. We fine-tune with QLoRA [4] on decisions distilled from a frontier teacher, compile to int4 ONNX, and

serve with Microsoft Foundry Local on a consumer RTX 2060 (6GB) of roughly signal-cabinet class. Control runs in SUMO on topologies built from measured London data (Euston A501 at DfT-measured peak-hour magnitude, a Bloomsbury grid, Old Street junction), with a MaxPressure [5] shield and anti-starvation floor between the model and the lights, and every decision bound into a hash-chained, signed audit record.

Contributions:

- **A documented QLoRA→int4→Foundry-Local chain.** To our knowledge the first published recipe for serving a custom fine-tuned model through Foundry Local, including the registration and execution-provider steps the toolchain does not document, with equal aggregate holdout accuracy between CPU and WebGPU serving.
- **A scale-saturation result.** Frontier distillation lifts a 0.6B student to 97–99% holdout agreement with the teacher, and a 3.8B student lands on the same numbers. For this task class, the bottleneck is not capability but the distillation ceiling.
- **A large, seed-unanimous effect and the measurement condition needed to see it.** Distillation is worth under two points on an oversaturated grid, but 18.3% on 30 of 30 seeds once that grid is calibrated to flow. Oversaturated test networks compress controller effect sizes, which has implications beyond this study.
- **The authorship finding.** Where delay effects are compressed, accuracy still decouples from delay, but fine-tuning raises the model’s share of logged decision ticks by up to 17 points and cuts its divergence from the classical reference four-fold. We define authored share on the decision-tick denominator, the quantity a decision audit needs, and separate it from proposal acceptance.
- **Honest nulls with mechanism.** We report the closed-loop nulls where they occur, the guard metrics that decide which delay numbers are comparable, and the failure tail with its audit-chain forensics.

II. RELATED WORK

LLMLight [1] distills GPT-4 rationales into LoRA-tuned students down to 0.5B and remains the closest prior work, but evaluates on CityFlow grids against cloud-served checkpoints. Traffic-R1 [2] applies R1-style RL to a 3B model

and CuraLight [3] is the one fine-tuning work on SUMO. DGLight [6] tunes an 8B model with GRPO and, usefully for calibration, reports the *entire* LLM-vs-MaxPressure gap at 1–2%. iLLM-TSC [7] inverts our topology by letting an RL agent propose and a cloud LLM veto, which prices the veto path at cloud latency and availability. Our deterministic local verifier costs $O(\text{phases})$ per decision. None of these serve on-device, none pair control with a tamper-evident audit, and none evaluate on measured UK topologies. Our shield is deliberately weaker than correct-by-construction shielding [8]: it composes preemptive action masking [9], post-posed default substitution, and an anti-starvation runtime monitor, and we discuss what a formal upgrade would require.

III. SYSTEM AND METHOD

A. Control loop and shield

Each junction is polled on a fixed decision interval (≈ 10 s simulated). The SLM receives junction state in one of four prompt formats of increasing privilege: *myopic* (per-phase halting counts), *sota* (delay-aware), *prediction* (arrival forecasts), *coordination* (neighbour notes). It must return exactly `{"phase": N}`. The shield masks the menu to conflict-free phases, substitutes the MaxPressure phase when output is invalid, and enforces a minimum-service floor. We define **divergence rate** as the fraction of SLM-authored decisions whose served phase differs from what MaxPressure would have chosen, and **authored share** as the fraction of logged decision ticks whose served phase originated with the SLM. Divergence is measured among decisions the SLM did author, so it describes how far the learned policy departs from the classical reference when it is in control. The harness field carrying it is named `override_rate` for historical reasons, which is a misnomer because the shield’s substitution path fires on 0.5% of decisions in the one cell where it fires at all. These are interference metrics in the sense of the shielding literature, not veto counts.

B. Audit

Every logged decision emits a record (halting-queue state, model proposal, shield proposal, served action, serving authorship, override flag) that is Ed25519-signed per junction and hash-chained into a single append-only chain per run, with model identity and latency recorded at run level, and chains verify end-to-end after each run. After a simulated collision the system assembles a cited evidence pack against a UK traffic-law knowledge base. That pack is deliberately not a verdict, following the legal-hallucination risk analysis of [10].

C. Distillation and training

We sample 48,137 junction states from eight training-reserved seeds, label them with a frontier teacher (cross-audited against a second: $\sim 98\%$ agreement), and render 59,533 examples in four formats with family-correct templates and completion-only loss. A hash-based split freezes a 5,580-example holdout pre-training. Students: Qwen3-0.6B and Phi-4-mini (3.8B), QLoRA $r=16$ NF4, merged to bf16, compiled

TABLE I
HOLDOUT ACCURACY (%) VS FRONTIER TEACHER, INT4-AS-SERVED.
STRICT-JSON COMPLIANCE IN PARENTHESES WHERE BELOW 100%.

	myopic	sota	prediction	coordination
Qwen3-0.6B stock	45.6	55.1 (93.4)	63.3	69.3
Qwen3-0.6B ft	87.9	97.7	97.6	99.0
Phi-4-mini stock	87.5	59.0	86.0	87.0
Phi-4-mini ft	88.7	97.1	97.4	99.3
Phi-4-mini LOTO	88.3	96.8	97.4	99.1

to int4 ONNX (RTN weight-only) via the `onnxruntime-genai` builder, registered in Foundry Local. The 0.6B trains on the serving GPU in 13.5 h, the 3.8B on an RTX 4090 in 2.3 h. Stock controls are the *same* base model on the *identical* int4 pipeline. Two 0.6B fine-tunes exist: Table I’s four-format generalist and the sota-only ft1 in all closed-loop sweeps. On ft1, CPU and WebGPU score identically in aggregate on the holdout (97.42% both; per-item outputs were not retained, so this is equal accuracy, not demonstrated output identity), so sweeps run GPU-served.

D. Closed-loop protocol

Closed-loop cells pair the controller against MaxPressure in loop and a fixed-time floor on the same seeds, disjoint from the eight training-state seeds. Because the route files are static, with fixed vehicle identities and departure times, and the SUMO seed varies microscopic car-following and lane-change randomness rather than the demand, that disjointness narrows trajectory-level leakage without closing it: head-to-head arm comparisons are unaffected, but generalisation beyond the trained demand is not established. Per topology and arm: $n=30$ seeds, paired t on mean network delay.

IV. EVALUATION

A. Offline: distillation saturates at 0.6B

Table I shows holdout agreement with the frozen teacher labels, stock vs fine-tuned, both students, int4-as-served.

Three observations. First, the two fine-tuned students are statistically indistinguishable despite a $6\times$ parameter gap, and both sit at the teacher-agreement ceiling (the myopic ceiling is intrinsically lower: the teacher saw state the myopic prompt hides). Distillation *saturates* this task at 0.6B. Second, where the gain lands depends on the base model: the 0.6B is lifted everywhere (+30 to +43 points), but stock Phi-4-mini is 100% format-compliant and 86–88% accurate on three formats, so it gains substantially only on the delay-aware format (+38.1 points). For a capable base model, fine-tuning raises decision quality, not format discipline. Third, a leave-out-topology student (LOTO, trained with no Old Street states) matches the generalist on every format, which is evidence against topology memorisation *in aggregate* rather than parity on the withheld layout: the holdout is scored whole with no topology filter, and Old Street is 4.2% of the sota rows and 0.4% of coordination, so this bounds a held-out-topology deficit near 7 points rather than measuring one. Per-topology accuracy was not retained.

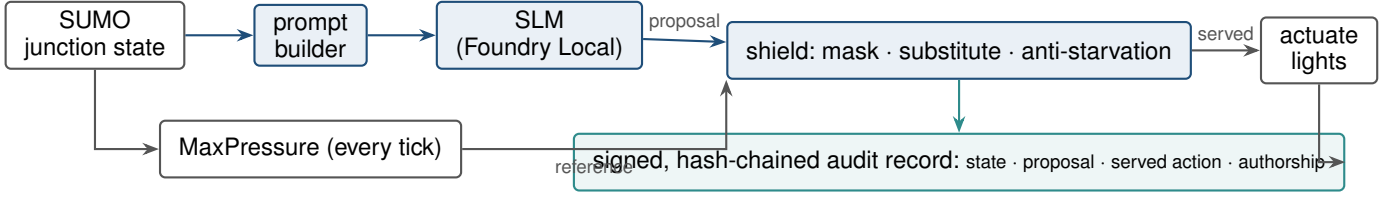


Fig. 1. The junction control loop. The deterministic path (grey) is complete without the SLM. The learned proposer (blue) attaches to it, and every served decision lands in the signed audit chain (teal).

TABLE II
CLOSED-LOOP MEAN NETWORK DELAY AGAINST BOTH CLASSICAL COMPARATORS ($n=30$ DISJOINT SEEDS). BOLD = SIGNIFICANT AT $p < .05$ ON THE FIXED-TIME COMPARISON.

Topology	Arm	vs fixed-time	vs MaxPr.	override
Euston peak-hour	ft	+15.5%	−3.9%	0.12
	stock	+7.6%	−6.2%	0.46
Bloomsbury grid	ft	+0.8%	+0.7%	0.17
	stock	+2.8%	+2.6%	0.46
Old Street	ft	−11.6%	−3.1%	0.11
	stock	−9.7%	−1.1%	0.60
Pooled ($n=90$)	ft	+1.6%	−2.1%	n/a

B. Closed-loop: delay decouples from accuracy

Table II reports the Qwen3-0.6B closed-loop sweep (fine-tuned ft1 vs identically-compiled stock ft0), paired per-seed against both classical comparators, which disagree in sign on one of three maps.

The comparator decides the outcome: against fixed-time every arm beats it on Old Street and nowhere else, while against MaxPressure only stock on the congested grid differs. What is invariant: a +42-point offline accuracy gain buys no consistent delay change, and the one protective effect is real. Stock harms the congested grid, +2.8% vs fixed-time ($p=.005$) and −1.8% head-to-head ($p=.022$). A Phi-4-mini confirmation arm (same 90 cells) lands arm-for-arm on the 0.6B student’s numbers, so saturation holds closed-loop as well.

The decisive evidence came from asking why the effects were so small. Throughput guards showed the Bloomsbury map never cleared: under 15% of vehicles completed and ~120 per run were teleported out of deadlock, and tripling the horizon made it worse (830 teleports), proving demand above capacity. We therefore demand-calibrated the map to 40% (87.6% completion, zero teleports, matching an earlier calibration of Old Street), kept the oversaturated cells, and re-ran all three arms on the identical 30 seeds. **On the network that flows, the stock model degrades delay by 11.9% [+10.0, +13.8] against the fixed-time floor while both fine-tuned students beat that floor by 8.8% (0.6B: −8.8% [−9.4, −8.2] and 3.8B: −8.8% [−9.3, −8.3]), drawing level with MaxPressure to within 0.03%. Head-to-head, fine-tuning is worth −18.3% [−19.8, −16.8], $p=2.0\times 10^{-21}$, winning 30 of 30 seeds, reproduced identically by the 3.8B stu-**

dent (−18.3%, 30/30, $p=8.2\times 10^{-22}$). The gridlock had been masking the effect: a deadlocked network has a ceiling on further degradation, and teleports truncate exactly the worst journeys, compressing good and bad control toward the same mean. The methodological corollary generalises: oversaturated test networks systematically understate controller effect sizes, and since the field evaluates mostly on synthetic grids under demand rarely reported in clearance terms, with its own calibration for the whole LLM-vs-MaxPressure gap at 1 to 2% [6], published effect sizes may be compressed by the same cause. Applying the same guard rule to the arterial corridor does not reproduce the effect. Calibrated Euston clears (completion rises from 45–51% to 68–74%), but the fine-tuned arm is still +8.5% against the fixed-time floor ($p=.12$) and the head-to-head gain shrinks to −3.7% in the mean, not significant ($p=.63$), with a −7.4% median on 23 of 30 seeds. We report that failure because the rule was applied where it could damage the claim, and it did. The title claim is therefore scoped, not withdrawn: authorship is what fine-tuning moves everywhere, and delay only where the network can clear, and not on every network that can.

Forensics on the failure tail qualifies this. Of 270 closed-loop cells, 14 are catastrophic ($\geq +20\%$ delay). All their audit chains verify, and the catastrophic seeds are baseline-easy or median realisations, so the failures are a controller property rather than seed hardness. The two fine-tuned students converge at the trajectory level on a substantial minority of cells: across 90 shared cells they produce bit-identical mean delay on nine and agree within 0.5 points on 25 more, differing on the remaining 56. The identical cells are separate runs whose wall-clock and median latency differ substantially but which serve the same phase sequence. Four of the five catastrophic fine-tuned cells fall inside this converged subset, which restates the convergence by construction rather than explaining it; the independent evidence is the decision-level agreement above. Two students of different architecture and $6\times$ different scale reproducing each other’s control trajectory exactly on a tenth of cells, and to within 0.5 points on a quarter more, is a behavioural counterpart to the offline saturation, though partial convergence is also consistent with both defaulting alike on easier cells. We did not separate those explanations.

C. Authorship: what fine-tuning actually moves

Three quantities must be kept apart, and conflating them is easy: the SLM’s share of *all* logged decision ticks (authorship), the fraction of its valid proposals that survive to

actuation (acceptance), and the fraction of its own decisions differing from MaxPressure (divergence). Fine-tuning moves all three by different amounts. On the three uncalibrated maps authorship rises from 0.39–0.62 to 0.56–0.70, acceptance from 0.54–0.68 to 0.70–0.90, and divergence falls from 0.46–0.60 to 0.11–0.17. On the calibrated grid the same contrast is sharper still, authorship 0.59 to 0.76–0.77 and divergence 0.56 to 0.04, which is where the 0.77 upper bound and 0.04 lower bound quoted in the abstract come from. The shield’s substitution path is inert on every cell but one (0.0% of decisions, 0.5% for the fine-tuned arm on Bloomsbury) because both stock and fine-tuned models emit well-formed proposals essentially always, so the competition for authorship is not the shield but the anti-starvation floor, which still forces 23–44% of decision ticks even under the best fine-tuned arm. This is the paper’s claim: for a shielded, audited edge controller, the deployable product of distillation is **authorship**. An accident audit over the stock Old Street deployment could attribute only 39% of decision ticks to the learned component, but over the distilled deployment it attributes 56%, with a deterministic timer authoring most of the balance. Causal accountability is the property regulators and investigators need, and that is exactly what moved, by 17 points.

D. Authorship cuts both ways

If fine-tuning makes the model author more of the decisions, it also makes the model’s inputs worth attacking. We ran an adversarial trust battery (honest / blatant / stealth neighbour misreporting) against stock and fine-tuned controllers at both scales. One scoping point governs every number below: trust gating was off in these runs, so the ledger altered no served phase. Detection is scored by replaying each run’s recorded claims against local sensing, which measures what the mechanism would have caught rather than what it prevented, and the delay costs are the attack’s effect on an undefended controller. Detection is controller-independent by design and in measurement, because the ledger scores neighbour claims against local sensing: a blatant liar is caught and collapsed under every controller (final trust 0.21–0.27 on the Bloomsbury grid and ≤ 0.03 on grid4 $\times$ 4, corroboration power stripped) while a calibrated stealth attacker evades everywhere (trust ≥ 0.92). But the delay cost of a blatant attack, relative to each controller’s own honest run, is usually *larger* for the fine-tuned arms on synthetic grids (up to +11 points against stock’s +7), though not uniformly. A controller that genuinely acts on its coordination inputs is moved further by poisoned ones. The trust ledger is therefore not a sufficient defence, and this study did not run a distilled deployment without one. The open problem is stealth misreporting, cheap and undetected.

We also measured a specialisation cost. On an adjacent structured task (vehicle-authorisation classification against a curated rule base), the fine-tuned students perform far *worse* than their own stock bases: the 0.6B student cannot emit the task’s {"class", "grant"} schema at all without in-prompt scaffolding (26/26 parse failures), having been captured onto {"phase": N}. A 200-item MMLU probe

shows only mild general-capability decline (57.5% $\rightarrow$ 53.0%, $p=.37$), so near-neighbour structured tasks are where the cost concentrates. This is an argument for architectural separation, not against distillation: the deterministic gate should keep authorisation, and the distilled model should keep phase selection.

E. Serving reliability at the edge

Sustained serving on consumer hardware is itself a finding: across the campaign we recorded three reliability findings. One long offline evaluation died at 5,500 of 5,580 rows on a host `OSError(28)` (no space left on device) and had to be re-run; registering builder-compiled models required undocumented synthesis of the inference manifest; and we observed the catalogue Qwen3-0.6B GPU artifact emitting malformed output after extended service uptime, which we record as an unlogged operational observation rather than a measured finding. All were mitigated in harness (endpoint rediscovery, reload-and-retry with intact statistics, fail-loud parse accounting), which also recovered two observed mid-run service self-restarts. Median decision latency is 0.94 s for the 3.8B student and 1.74–1.75 s for the 0.6B students, so the larger model is the faster arm to serve, a property of the compiled artifacts rather than parameter count. Both sit comfortably inside the 10 s interval, though one individual call in the 0.6B arms exceeded it (13.8 s, against a next-worst 9.9 s), so a deployment needs a hard timeout with deterministic fallback.

V. LIMITATIONS

Simulation-only, three topologies, one city. Neither student improves on the fixed-time floor on the saturated arterial, where both are worse than it, and the trust mechanism does not stop a stealth attacker. We distil labels, not reasoning traces, and Orca-style trace distillation [11] might transfer more. Our myopic format shows the cost of privileged teachers directly. The shield is a heuristic composite, not a synthesized shield with correctness guarantees [8]. “Lossless” int4 is scoped strictly as equal aggregate holdout accuracy under weight-only RTN, with per-item outputs not retained. A seeded 200-item MMLU retention probe through the same endpoints finds no significant general-capability loss at 3.8B (57.5% stock vs 53.0% fine-tuned, $p=.37$). At 0.6B the stock model already scores at chance on this format, so forgetting is unmeasurable there by floor effect.

VI. CONCLUSION

A 0.6B SLM, distilled from a frontier teacher for 13 GPU-hours on the same 6GB device that serves it, reaches the teacher-agreement ceiling, and six times the parameters add nothing. Closed-loop it beats a fixed-time floor where the network can clear: 8.8% on the calibrated grid and 11.6% on the most complex geometry. Elsewhere it matches that floor, and on a saturated arterial no arm, distilled or not, improves on it. Beyond delay it authors more of what the junction does: its share of logged decision ticks rises by

up to 17 points as the fairness floor is needed less, its divergence from the classical reference falls four-fold, and the stock model’s measured harms disappear. For edge traffic control, where every actuation must survive an audit, fine-tuning’s contribution is not delay but that shift in attributable authorship. Future work: GRPO-style closed-loop tuning [6], formal shield synthesis, and a hardware-in-the-loop pilot.

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